

DHARUNKUMAR C

Python GenAI Engineer — Generative AI — LLMs — RAG — AI Applications

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PROFESSIONAL SUMMARY

IT graduate with hands-on experience building **Python-based Generative AI and machine learning applications** using LLMs, RAG, embeddings, semantic search, LangChain, LangGraph, and cloud AI services. Experienced in developing **production-oriented AI workflows** for fraud analysis, healthcare claim adjudication, document processing, retrieval, decision support, and conversational applications. Skilled in building **modular, API-driven applications** using FastAPI and working with vector databases and enterprise data stores. Experienced in **AI evaluation, observability, monitoring, guardrails, auditability, access control, and human-in-the-loop workflows**. Strong foundation in software engineering, machine learning, computer vision, data processing, and **production-grade AI application development**.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript

Generative AI & LLMs: Generative AI, LLMs, RAG, Prompt Engineering, Tool Calling, Semantic Search, Embeddings, Conversational AI, Context Management, AI Agents

GenAI Frameworks: LangChain, LangGraph, LlamaIndex, Hugging Face

AI Platforms & Models: AWS Bedrock, Azure OpenAI, Llama 3, Claude, Amazon Titan Embeddings

Vector Databases & Data Stores: FAISS, OpenSearch Vector Database, PostgreSQL, DynamoDB, Redis, Azure Blob Storage, Amazon S3

Backend & APIs: FastAPI, Flask, Streamlit, REST APIs, API Design, Microservices

AI Evaluation & Observability: MLflow, RAGAS, Prometheus, Grafana, Model Evaluation, Prompt Evaluation, Monitoring

Cloud & Deployment: AWS Lambda, Amazon SQS, API Gateway, IAM, Docker, Kubernetes

AI Governance: Guardrails, Human-in-the-Loop, Access Control, Audit Logging, Traceability, Responsible AI

Machine Learning: Machine Learning, Deep Learning, Computer Vision, Image Processing, Object Detection, Image Segmentation

Tools: Git, Postman, Jupyter Notebook, VS Code

EDUCATION

Parul University, Vadodara

07/2022 – 06/2026

Bachelor of Engineering in Information Technology

PROFESSIONAL EXPERIENCE

WNS Global Solutions

12/2025 – 06/2026

Generative AI Intern

- Contributed to AI/ML solutions involving data processing, validation, structured analysis, and quality-focused workflows.
- Built document ingestion and preprocessing workflows using **Python and LangChain** for enterprise AI applications.
- Implemented semantic retrieval using **FAISS** for document and business information retrieval.
- Processed claim and transaction data using **PostgreSQL**, applying validation and accuracy-checking workflows.
- Implemented AI-assisted workflows for business information retrieval, summarization, missing-information identification, and decision support.
- Designed and tested fraud-analysis workflows involving transaction patterns, customer information, compliance rules, and risk indicators.
- Evaluated AI responses for consistency, relevance, quality, and accuracy across business workflows.
- Maintained audit logs and monitoring workflows to support traceability, quality, and consistency.

Banking Fraud Alert & Risk Scoring AI Platform

- Developed a GenAI-powered fraud investigation and risk scoring platform using **Python, FastAPI, LangGraph, and AWS Bedrock**.
- Designed an event-driven multi-agent workflow with agents for fraud classification, risk scoring, rule validation, customer profiling, and case recommendation.
- Implemented **RAG** using Amazon Titan Embeddings and OpenSearch Vector Database to retrieve AML rules, fraud patterns, compliance policies, and regulatory documents.
- Integrated Bedrock Claude for fraud investigation summaries, risk explanations, suspicious activity narratives, and recommendations.
- Built FastAPI endpoints for integration with internal banking applications and enterprise workflows.
- Implemented audit logging and monitoring using DynamoDB and CloudWatch for traceability of AI-driven decisions.
- Added guardrails for unsafe recommendations, hallucinated compliance statements, and unauthorized data exposure.
- Evaluated RAG workflows using **RAGAS** metrics including faithfulness, answer relevance, context precision, and

context recall.

Healthcare Claim Adjudication GenAI Assistant

- Designed and implemented a healthcare claim adjudication assistant using **Python, LangChain, RAG, Azure OpenAI/Llama 3, and FAISS**.
- Built document ingestion workflows for claim forms, medical bills, discharge summaries, policy documents, and adjudication guidelines.
- Implemented embedding-based retrieval for policy clauses, member eligibility, CPT/ICD descriptions, medical guidelines, and adjudication rules.
- Developed prompts for claim summaries, approval/denial reasoning, missing-document alerts, and adjudication recommendations.
- Implemented conversational memory using **Redis** to support multi-turn claim analysis.
- Used PostgreSQL for claim metadata, member information, claim status, and validation workflows.
- Implemented Celery-based batch processing for large-volume claim documents.
- Used **MLflow** for prompt versioning, model outputs, and evaluation workflows, with Prometheus/Grafana for monitoring latency, success rate, and response quality.

ACADEMIC PROJECTS

Brain Tumor Detection

- Built an MRI image classification application using **VGG16** to identify brain tumor categories.
- Performed image preprocessing, augmentation, and feature extraction to prepare MRI data for model training.
- Developed and evaluated the deep learning model using **TensorFlow and Keras** for image classification and prediction.
- Created a **Flask** interface for image upload and prediction, integrating the trained model into an application workflow.
- **Technologies:** Python, TensorFlow, Keras, VGG16, Flask, OpenCV, NumPy, Pandas, Matplotlib

Steel Surface Defect Detection

- Developed a computer vision system for detecting and classifying defects in steel surface images.
- Performed image preprocessing and visual data preparation to improve the quality of input images.
- Implemented defect classification and segmentation-based location analysis using deep learning techniques.
- Evaluated prediction results for accuracy and consistency to support automated visual inspection.
- **Technologies:** Python, Deep Learning, Computer Vision, Image Processing, PyTorch, Image Segmentation

ADDITIONAL INFORMATION

- Available for rotational shifts, including night shifts, weekends, and holidays.
- Strong attention to detail and accuracy in repetitive data review and validation workflows.
- Interested in Generative AI, LLM applications, RAG systems, AI agents, machine learning, and data quality.
- Comfortable working with high-volume data processing, validation, and structured analysis.

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate
- Machine Learning with Python – Great Learning Academy
- Microsoft Azure AI Fundamentals (AI-900)
- Google Generative AI Fundamentals
- NPTEL – Computer Networks And Internet Protocol
- NPTEL – Theory of Computation
- NPTEL – Introduction to Internet of Things

LANGUAGES

English – Fluent — Hindi – Fluent — Tamil – Fluent — Telugu – Professional Proficiency